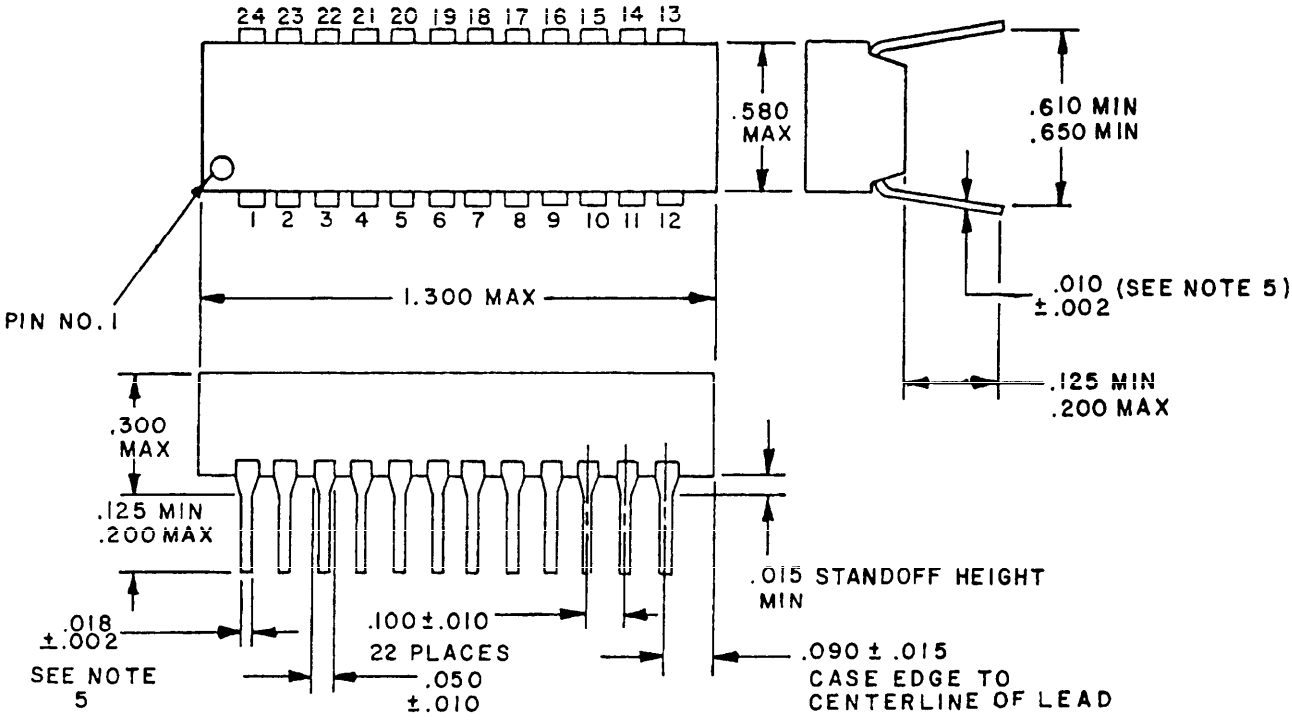
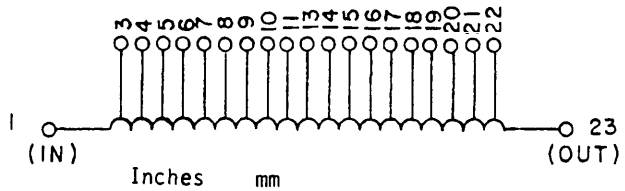
MILITARY SPECIFICATION SHEET

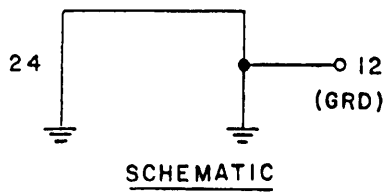
DELAY LINES, 20-TAP, DUAL-IN-LINE, 24-PIN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the delay lines described herein shall consist of this specification and the latest issue of MIL-D-83531.







SCHEMATIC

Inches	mm
.002	0.05
.010	0.25
.015	0.38
.018	0.46
.050	1.27
.090	2.29
.100	2.54
.125	3.18
.200	5.08
.280	7.11
.300	7.62
.580	14.73
.610	15.49
.650	16.51
1.300	33.02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Location and shape standoffs are optional. Standoff height is required.
4. Pin 2 is not connected.
5. Terminal dimensions are prior to hot solder dip.
6. Schematic is for general information only.

FIGURE 1. Case and mounting dimensions.

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REQUIREMENTS:

Operating temperature: -55°C to +125°C.

Weight: 6.0 grams maximum.

Case material: Plastic.

Terminal material: Type LR(0.52) G-35-C* in accordance with MIL-STD-1276. Prior to shipment, terminals shall be hot solder dipped in SN60, SN62, or SN63 in accordance with QQ-S-571 to bottom of standoffs, minimum.

Design and construction: See figure 1.

Total delay time: See table I.

Rise time: 25% of total delay, maximum.

Voltage attenuation: See table I.

Distortion: ±10% at maximum delay.

Delay time variation with temperature: 100 PPM/°C maximum.

DC resistance (RDC): $RDC = \text{impedance}\left(\frac{\text{attenuation}}{1 - \text{attenuation}}\right)$

EXAMPLE: $RDC = 50\left(\frac{0.03}{1-0.03}\right)$ $RDC = 1.5$ ohms

Nominal characteristic impedance: See table I.

Insulation resistance: In accordance with MIL-D-83531.

Input pulse:

Peak pulse voltage: 1 to 3 volts ±5%.

Pulse width: Three times nominal total delay.

Duty cycle: 25%.

Salt spray (corrosion): In accordance with MIL-D-83531.

Thermal shock: In accordance with MIL-D-83531.

Shock: In accordance with MIL-D-83531.

Tap accuracy and total accuracy:

Total delay 0 to 10: ±0.5 ns.
Total delay 11 to 25: ±1 ns or 5% (whichever is greater).
Total delay 26 and up: ±2 ns or 5% (whichever is greater).

Part number: The part number shall be as follows:

EXAMPLE:

M	83531/03	-001
Military designator	Specification sheet number (note zero)	Dash number (see table I)

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TABLE I. Part numbers and electrical measurements.

Part number M83531/03-	Nominal characteristic impedance (Ω) ±10 percent	Total delay (ns) ±5 percent	Maximum attenuation (percent)
001	50	50	6
002	50	60	7
003	50	80	9
004	50	100	11
005	50	150	11
006	50	200	12
007	100	50	6
008	100	60	6
009	100	80	6
010	100	100	7
011	100	150	7
012	100	200	8
013	100	300	10
014	100	400	11
015	200	40	3
016	200	80	4
017	200	120	5
018	200	200	6
019	200	300	6
020	200	400	7
021	200	500	8
022	200	600	10
023	200	800	16
024	250	50	3
025	250	100	4
026	250	150	5
027	250	200	8
028	250	250	8
029	250	300	8
030	250	400	9
031	250	500	11
032	250	600	13
033	250	800	14
034	250	1,000	16
035	500	200	4
036	500	300	7

See footnote at end of table.

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TABLE I. Part numbers and electrical measurements - Continued.

Part number M83531/03-	Nominal characteristic impedance (Ω) ±10 percent	Total delay (ns) ±5 percent	Maximum attenuation (percent)
037	500	400	7
038	500	500	8
039	500	600	9
040	500	800	14
041	500	1,000	17
042	500	1,200	18
043	500	1,500	21
044	500	2,000	24

1/ Delay between successive taps shall be 1/20 of the nominal total delay.

QUALITY ASSURANCE PROVISIONS:

Extent of qualification: The extent of qualification shall be as specified in MIL-D-83531, for example:

Qualification and testing of M83531/03-001 and M83531/03-044 shall be sufficient to grant qualification to M83531/03-001 through M83531/03-044 inclusive.

Supersession data: MIL-D-83531/3 supersedes DESC drawing 85012 when a qualifiec source is available.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5999-0170-03)

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